

KAVI MOOKHERJEE AMODT

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EDUCATION AND COURSEWORK

UC Berkeley

Until 2026

Computer Science, Cognitive Science; Linguistics Minor

3.72 GPA

CS: Probability and Random Processes (EECS 126), Artificial Intelligence (CS 188), Algorithms (CS 170), Computer Security (CS 161), Internet Architectures and Protocols (CS 168), Algorithmic Economics (CS 177), Computability and Complexity (CS 172), Econometrics (Econ 141), Computer Architecture (CS 61C), Data Structures (CS 61B), Discrete Math and Probability (CS 70), Linear Algebra (CS 56), Calculus (51/52), Statistics (2)

Cognitive Science and Linguistics: Courses in psychology, neuroscience, data science, and linguistics.

EXPERIENCE

Software Engineering Intern

2024

Tools for Humanity

San Francisco, CA

- Wrote Rust, Python, and Bash for an iris scanning device called the Orb.
- Built Rust CLI tools for accessing Orb metrics.

Research Assistant

2025–

Language and Cognitive Development Lab

UC Berkeley

- Built experiment for word-object learning models using dynamic field theory in MATLAB.
- Wrote data processing scripts in Python.

Research Assistant

2021–2023

UCSC Linguistics Department

Santa Cruz, CA

- Eyetracking research on underspecification, parsing of syntactic ambiguities.
- Operated eye-tracker, worked with participants, collected data.
- Worked with Amanda Rysling (UCSC), Jack Duff (UCLA).

PROJECTS

Secure File Sharing System

- Designed and implemented a cryptographically secure file sharing system in Go.
- Built from scratch with no vibe coding. 2000+ lines.
- Tested in a fully adversarial server environment where adversaries have full write access to datastore and read access to keystore.

Prediction Market Simulator

- Simulating and visualizing prediction markets with LMSR online learning algorithm in C++.

Digit Classifier

- Implemented a neural network inference engine in RISC-V assembly, capable of classifying handwritten digits through matrix operations and ReLU activation.

Wordle Bot

- Built an entropy-maximizing Wordle solver in Python that selects optimal guesses by maximizing Shannon information gain.

SKILLS

Languages: Python, Go, C++, C, Java, R, RISC-V Assembly, SQL, Bash

Libraries & Frameworks: TensorFlow, PyTorch, NumPy, SciPy, pandas, Matplotlib

Tools & Platforms: Git, Jupyter, Linux, Model Context Protocol (MCP)

Domains: Machine learning, applied cryptography, information theory, probabilistic modeling, computer architecture